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School of Science Engineering and Technology Skills

Department of Engineering

Subject – AWS Lab

Experiment – 7

Title: Working with NoSQL Databases using Amazon DynamoDB

Introduction:

Amazon DynamoDB is a fully managed NoSQL database service that provides fast and predictable performance with seamless scalability. Unlike relational databases, DynamoDB is schema-less (except for the primary key), making it ideal for high-scale applications, mobile backends, and gaming.

Key Concepts:

- Tables: Collections of data.
- Items: Similar to rows in RDBMS; a group of attributes.
- Attributes: Similar to columns; the smallest unit of data.
- Primary Key: Can be a Partition Key (Simple) or a Partition Key + Sort Key (Composite).

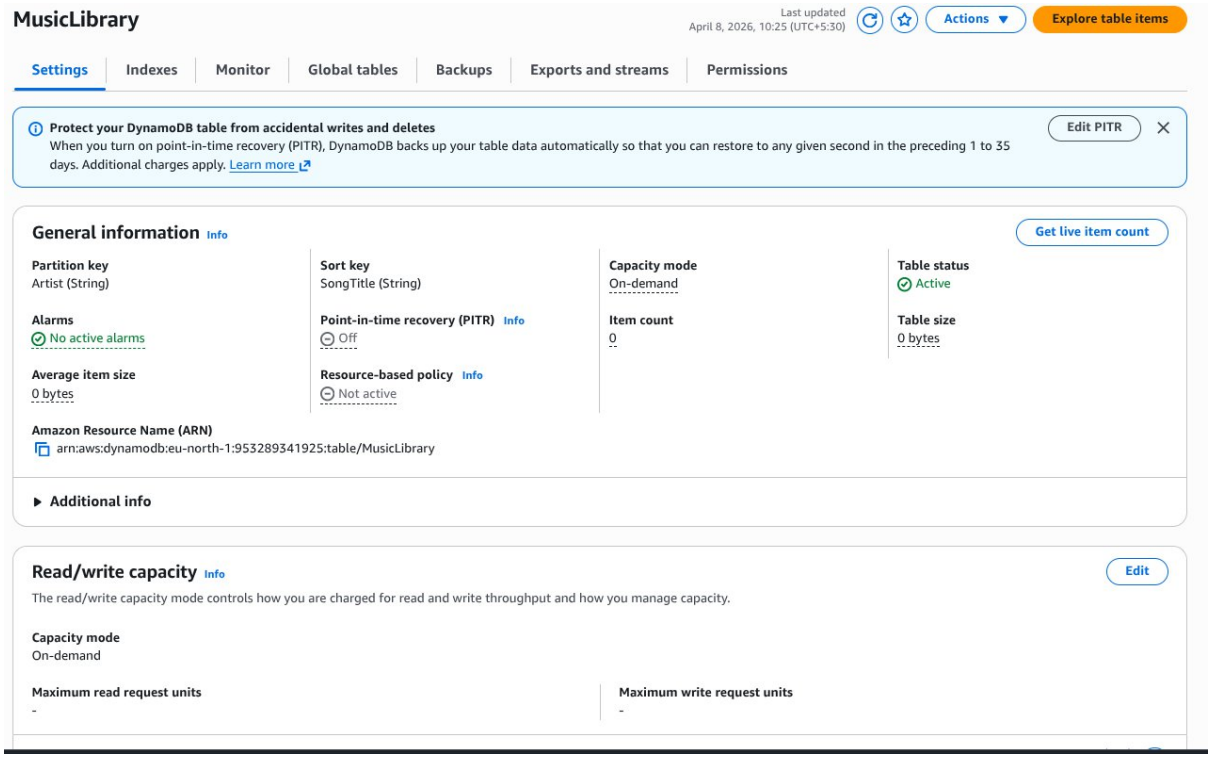
Lab Objective:

1. Understand the core architecture and scaling properties of DynamoDB.
2. Create a DynamoDB table with a defined Primary Key.
3. Perform CRUD (Create, Read, Update, Delete) operations using the AWS Management Console.
4. Explore the "Scan" vs. "Query" functionality.

Part 1: Launching a DynamoDB Table

Step-by-Step Instructions:

1. Navigate to DynamoDB: Search for "DynamoDB" in the AWS Console.
2. Create Table: Click the Create table button.
3. Table Details:
    - Table name: MusicLibrary
    - Partition key: Artist (String)
    - Sort key: SongTitle (String)
4. Table Settings:
    - Keep Default settings (this selects "Provisioned" capacity with Auto Scaling).
5. Finalize: Click Create table. Wait for the status to change from Creating to Active.



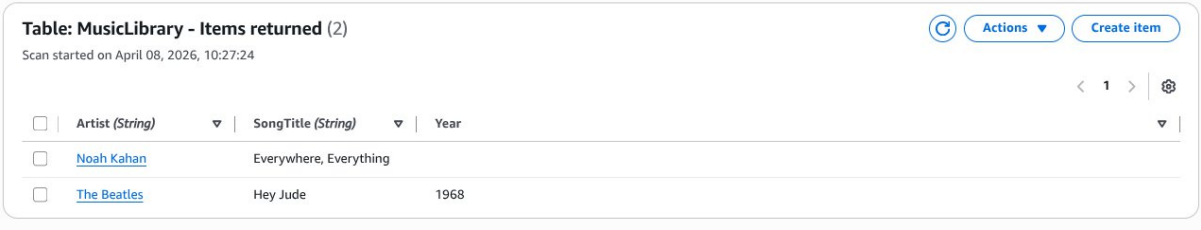
Part 2: Performing CRUD Operations

CRUD operations allow you to manage the lifecycle of your data.

A. Create (Put Item)

1. Select the MusicLibrary table and click **Explore table items**.
2. Click **Create item**.
3. **Attribute Values:**
    - Artist: The Beatles
    - SongTitle: Hey Jude
4. **Add New Attribute:** Click **Add new attribute** > **Number**.
    - Attribute name: Year | Value: 1968.

5. Click **Create item**. Repeat this to add 2-3 more items.

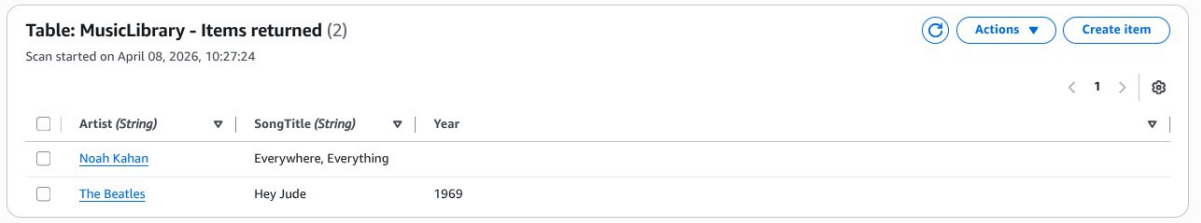


B. Read (Get Item)

1. In the **Items returned** list, you can see all entries.
2. To view details of a specific item, simply click on the Partition Key value.

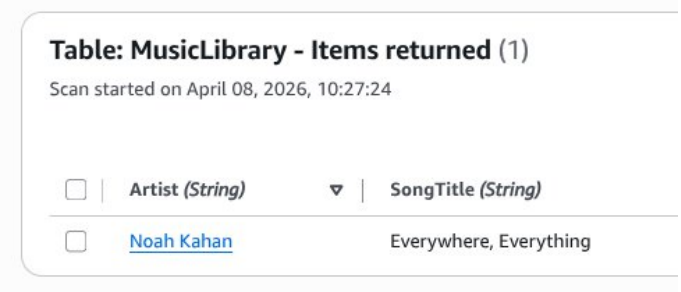
C. Update

1. Select an existing item (e.g., Hey Jude).
2. Change the Year attribute from 1968 to 1969.
3. Click **Save changes**.



D. Delete

1. Select the checkbox next to an item you wish to remove.
2. Click **Actions** > **Delete items**.
3. Confirm the deletion.

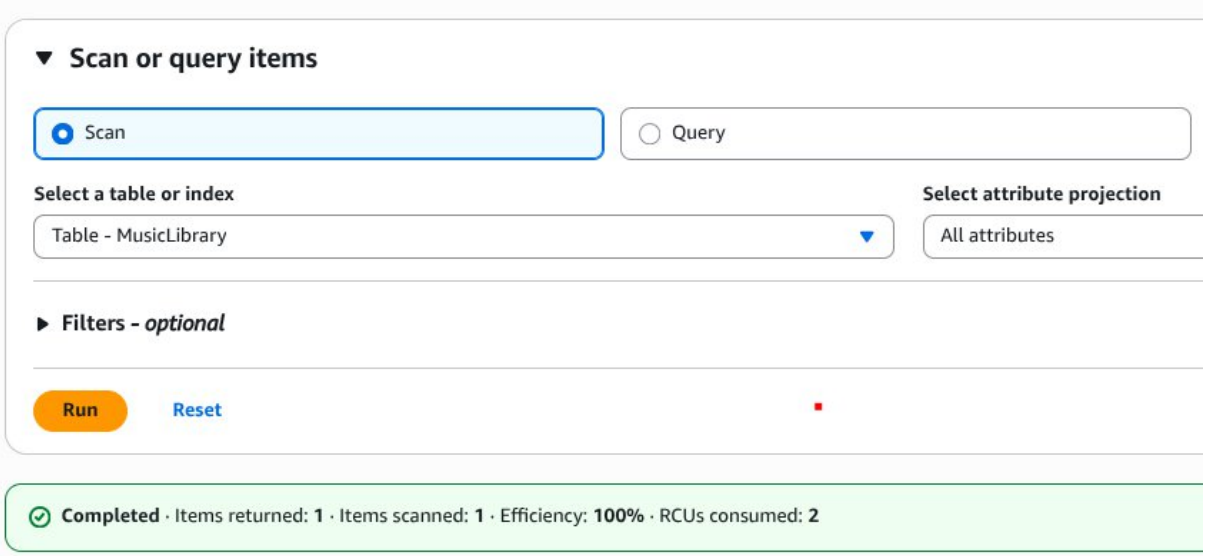


Part 3: Querying vs. Scanning

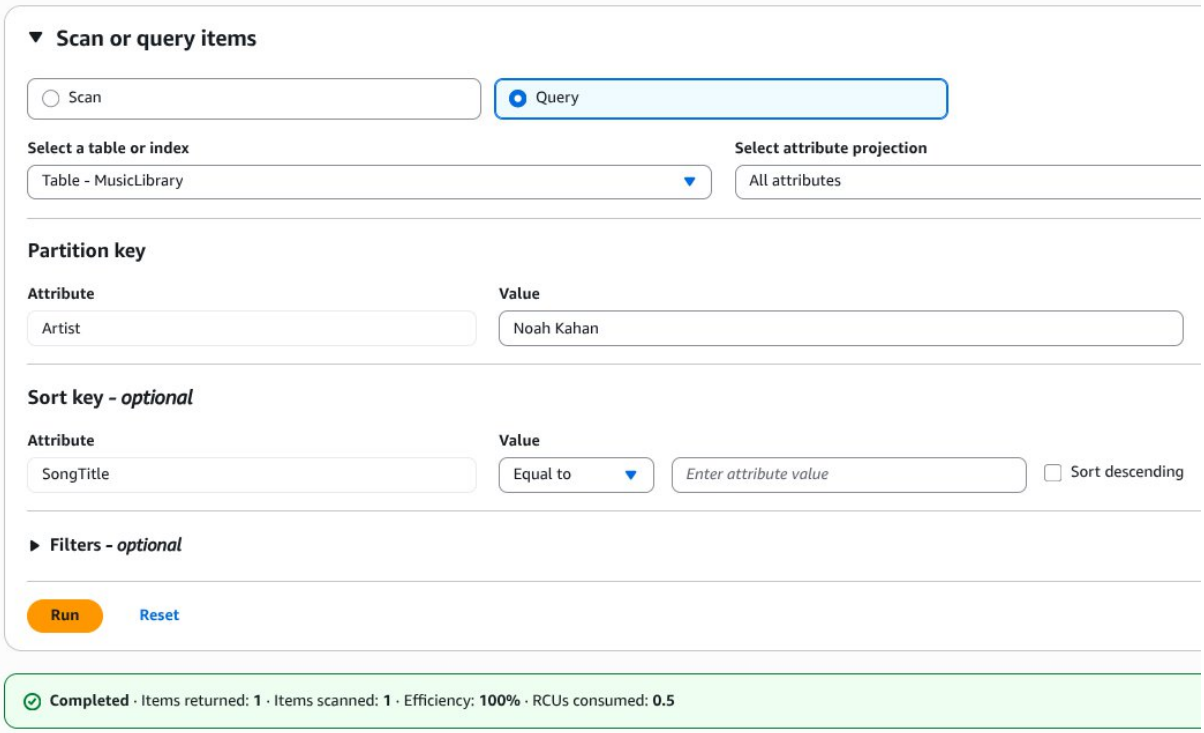
Efficiency in NoSQL depends on how you retrieve data.

Step-by-Step Instructions:

1. Under **Explore items**, toggle the search between **Scan** and **Query**.
2. **Scan:** Leave the filters empty and click **Run**. Notice it looks at every item in the table (expensive for large tables).



3. **Query:** * Select **Query**.
    - Enter a specific **Partition key** (e.g., The Beatles).
    - Click **Run**. Notice it only searches the partition for that specific artist (much faster and cheaper).



Observation Table:

Operation	Input Data	Outcome/Result
Table Creation	Name: MusicLibrary	Table status set to Active
Create Item	Artist: ArtistName	Item visible in the table
Update Item	Change Year	Value updated in JSON view
Query	Filter by Artist	Only specific artist items returned

Conclusion

In this lab, you successfully provisioned a NoSQL database, managed data through CRUD operations, and analyzed the performance differences between scanning and querying. This foundation is essential for building serverless, scalable cloud applications.